

Date: 2026-05-14
Request ID: 100048528

Microwave				
Element(s)	Al, Zr			
SOP#				
Acid Cocktail	9 mL HCl, 3 mL HNO3, 3 mL H2SO4			
Rack	7			
Vessel	PTFE			
Stir	90%			
Time (min)	Power (W)	T1 (°C)	T2 (°C)	P (bar)
15	1200	180	65	110
15	1500	240	65	110
30	1500	240	65	110
sample(s)	weight (g)	volume (mL)		
200127910_1	0.1063	250		
200127910_2	0.1053	250		
200127911_1	0.1087	250		
200127911_2	0.1054	250		

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Al ICP analysis

sample	Dilution	conc. [mg/L]	Al_ppm	%Al
200127910_1	1/1		0.00	50.31
200127910_2	1/1		0.00	50.31

al result: 0.00 ppm

al result: 50.31 %

al oxide factor: 1.889

al oxide result: 95.04 %

al lot: *Manufacturer: Assurance, 27-171ALT exp: 2026-07-30*

Zr ICP analysis

sample	Dilution	conc. [mg/L]	Zr_ppm	%Zr
200127910_1	50/6	1.813	35532.30	3.55
200127910_2	50/6	1.917	37927.35	3.79

zr result: 36729.82 ppm

zr result: 3.67 %

zr oxide factor: 1.351

zr oxide result: 4.96 %

zr lot: *Manufacturer: Assurance, 28-64ZRY exp: 2026-10-30*

Note(s): *Didn't recover all of alumina from both microwave and HF digestion so reported Al value based on mass balance.*

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Al ICP analysis

sample	Dilution	conc. [mg/L]	Al_ppm	%Al
200127911_1	1/1		0.00	50.17
200127911_2	1/1		0.00	50.17

al result: 0.00 ppm

al result: 50.17 %

al oxide factor: 1.889

al oxide result: 94.77 %

al lot: *Manufacturer: Assurance, 27-171ALT exp: 2026-07-30*

Zr ICP analysis

sample	Dilution	conc. [mg/L]	Zr_ppm	%Zr
200127911_1	50/6	2.030	38906.78	3.89
200127911_2	50/6	1.943	38405.28	3.84

zr result: 38656.03 ppm

zr result: 3.87 %

zr oxide factor: 1.351

zr oxide result: 5.22 %

zr lot: *Manufacturer: Assurance, 28-64ZRY exp: 2026-10-30*

Note(s): *Didn't recover all of alumina from both microwave and HF digestion so reported Al value based on mass balance.*

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A] - [C-A]) / (B-A) * 100\%$$

$$ppm\ M+ = [conc.][volume][dilution] / [weight]$$

$$\%M+ = [conc.][volume][dilution] / ([weight] * 10,000)$$

$$\%MO = [oxide\ factor] * \%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1 - (\%LOI) / 100])$$

$$\%M+ \ [dried] = \%M+ / ([1 - (\%LOI) / 100])$$

$$\%MO \ [dried] = \%MO / ([1 - (\%LOI) / 100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS]) / ([weight])$$

$$\%Cr_2O_3 = 1.462 * [\%Cr(VI)]$$

$$\%Cr(III) = total_Cr - Cr(VI)$$